GIS or GPS? A Comparison of Two Methods For Assessing Route Taken During Active Transport

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Introduction: Current methods of assessing routes taken during active transport rely on subjective recall of trip length and barriers encountered enroute or the utilization of objective measures (Geographic Information Systems –[GIS]) that may not represent actual travel patterns. This study examined the utility of Global Positioning Systems (GPS) to measure actual routes taken compared with GIS-estimated travel distance and barriers encountered.

Methods: Comparisons between GPS and GIS routes were performed for 59 of 75 children who wore a GPS during the journey to school on a single occasion. Home and school addresses were reported by parents and geocoded in GIS. Children were provided with a GPS and were instructed to travel their normal route to and from school. Data were collected between March and November 2005 and exported to the GIS to determine travel distance, number of busy streets crossed, and the ratio of busy streets to the total streets traveled on. Data analysis was performed in August 2006.

Results: No differences were observed between GPS-measured journeys to and from school on any of the examined variables. No differences were observed between GIS and GPS measures of travel distance ($p>0.05$). GIS-estimated travel routes crossed a significantly ($p<0.05$) higher number of busy streets (GIS: 1.68±0.12 vs GPS: 1.19±0.11) and traveled on a higher ratio of busy streets to total streets traveled on (GIS: 0.46±0.03 vs GPS: 0.35±0.04) ($p<0.05$) compared with GPS-measured actual travel routes.

Conclusions: Geographic Information Systems provides estimates of travel distance similar to GPS-measured actual travel distances. Travel routes estimated by GIS are not representative of actual routes measured by GPS, which indicates that GIS may not provide an accurate estimate of barriers encountered. The continued use of GPS in active transport research in encouraged.

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Introduction

Our understanding of the barriers to walking and cycling (active transport) to and from school is limited by an over-reliance on self-report or Geographic Information System (GIS) measurement of travel paths,[1–3] which may not accurately reflect actual travel paths. Previous research has demonstrated that Global Positioning Systems (GPS) provide an acceptable alternative to GIS when measuring distance traveled during walking and cycling[4] and can be used to track activity in outdoor environments.[5] GPS may be useful in tracking active transport through urban environments and providing accurate measures of travel distance. Therefore, the purpose of this research was to determine if (1) journey-to-school distance is comparable when measured by GPS and GIS, (2) differences exist in the number of busy streets crossed enroute to school as measured by GPS and GIS, and (3) if travel paths measured by GPS and GIS are similar.

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Methods

Participants

Data were collected from a convenience sample of 75 primary school children (mean age, 11.09 ± 0.79 years) who were recruited as part of a larger study and who participated in any parent-recalled active transport journey to or from school in Rockhampton, Australia. Written and informed consent was provided by all students and their parents/guardians, and ethical clearance was granted by Central Queensland University's Human Research Ethics Review Panel. Participating children were required to wear a GPS (Garmin etrex, Garmin Ltd., Olathe KS) for a single journey to and from school, and the majority of the sample (73.2%) engaged in more than three journeys to or from school in a usual week.

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Table 1. Journey distances, number of busy streets crossed, and busy street ratios during school journeys as measured by GIS and GPS

	GPS journey to school			GIS journey to school		
Variable	***n***	**Mean (SE)**	**Range**	**Mean (SE)**	**Range**	***p***
Distance (meters)	59	1081.58 (109.39)	4360.00	1067.41 (98.18)	3060.00	0.747
Busy streets (No.)	59	1.19 (0.11)	4.00	1.68 (0.12)	3.00	<0.001
Busy street Ratio	59	0.35 (0.04)	1.00	0.46 (0.03)	0.83	<0.001

GIS, Geographic Information Systems; GPS, Global Positioning Systems; SE, standard error.

Protocol

Students met with the primary investigator for a familiarization session on correct operation, placement of the GPS, and purpose of the study before data collection commenced and also received an information sheet detailing unit operation. Students wore the GPS on manufacturer-provided lanyard.

Students were required to wait at the journey origin until the GPS units displayed a message indicating that sufficient satellite signals had been acquired (there was a minimum of four satellites). This generally required the students to remain stationary for approximately 1 minute prior to journey commencement. If the message did not appear, the students were instructed to repeat the process two additional times. If unsuccessful after three attempts, students completed their journey and were provided with a replacement unit to repeat the data collection at another time.

Measures

Participant residential and school addresses were geocoded using Rockhampton City Council's GIS database using Genamap version 8.1.1 (Genaware, Ltd, Sydney, 2004). Route taken and distance traveled were calculated using GIS shortest route analysis between the student's home and school [2,6] and was limited to designated roadways to determine route and travel distance, and was not permitted to cross nature reserves, parks, or informal paths. Assessment of active transport, GIS, and GPS travel paths was conducted between March and November 2005, and analysis was performed in August 2006.

The Garmin etrex is a small unit (111.8 × 50.8 × 30.5 mm) capable of providing estimates of travel distance to within 2% of the actual distance.[4] Data were downloaded to MapSource Metroguide (MapSource Metro Guide, Sydney, Australia, 2002) at the completion of each child's data collection. Following mapping of the route taken by GPS and GIS, measures of distance traveled, the total number of streets and busy streets traveled on, the number of parks, and busy streets crossed were determined manually for both GPS and GIS. Busy streets were defined as streets holding a speed limit of ≥60 km/hr,[7] which reflect a higher traffic volume on these streets compared with other residential streets.[8] A ratio of busy streets to total streets traveled was created as a measure of dissimilarity between routes taken measured by GPS and GIS. Students whose GPS tracks began outside of a 20-m buffer of their journey origin and ceased outside of the buffer area at the journey destination (n=4) were excluded from analysis leaving 71 for analysis. These tracks were excluded, as tracks beginning outside of the 20-m buffer areas may provide over-estimations of distance traveled.[4] Journeys to school were not significantly different from journeys from school on any variable examined (p>0.05), therefore, the journey to school was used for all comparisons to GIS.

Results

Of the 71 cases, data on both the journey to and from school were available for 42 cases, whereas 59 and 54 cases were available for journeys to and from school, respectively. Variation in complete cases is due to tracks including signal dropout being removed from analysis, as they represent incomplete monitoring periods. GIS estimates of journey distance were not significantly different from GPS estimates of distance (p=0.747), however, GIS estimates revealed significantly higher numbers of busy streets crossed (p=<0.001) and busy street ratios (p=<0.001) than measured by GPS routes (Table 1). Figure 1 provides an example of routes

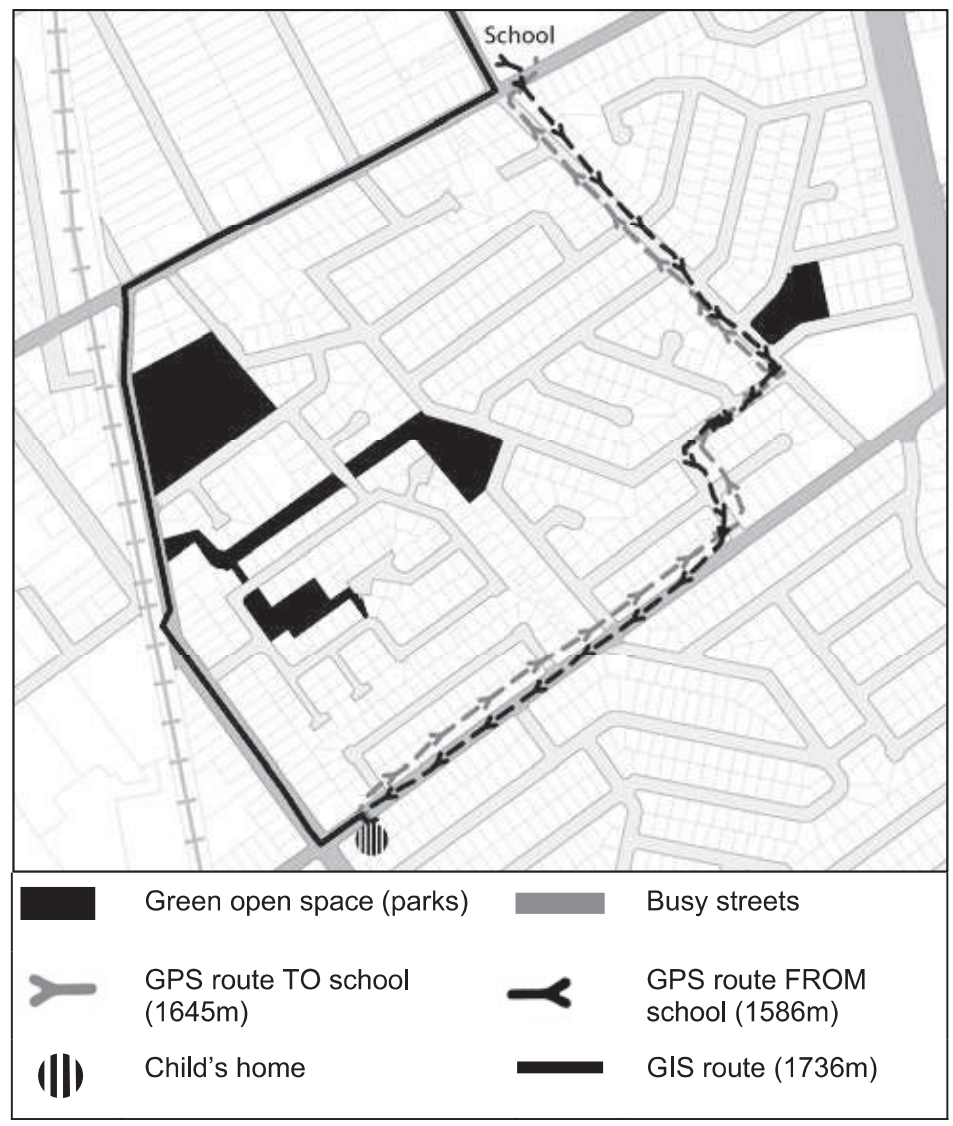

Figure 1. Illustration of GIS and GPS measured routes to and from school that differ on the number of total and busy streets travelled on.

measured by GPS and GIS that differ on the number of total and busy streets traveled on, yet provided similar measures of travel distances via different routes.

Discussion

This study supports the usefulness of GPS for monitoring physical activity, specifically active transport. Results show that although GIS and GPS provided comparable measures of distance, GIS failed to consider route choice, an important determinant in the decision to engage in active transport. Our findings suggest that GIS and GPS can provide different travel route information, as evidenced by the higher number of busy streets crossed and busy street ratios measured by GIS than by GPS. This suggests that children seek out routes that, although of similar distance, are of lower traffic volumes (as the definition of busy streets used is reflective of higher traffic volumes on these streets).[8] GIS may, therefore, provide misrepresentations of the route taken as well as frequency and type of barriers encountered. However, depending on the research requirements, self-report, GIS, and GPS may all be useful in examining distance traveled and barriers encountered. Self-report is a cost-efficient method in large populations, although the accuracy of estimates is unknown and subject to the cognitive abilities of participants. GIS provides similar estimates of travel distance to GPS, but estimation of routes and associated barriers may not represent actual travel patterns. Future research may wish to permit GIS shortest route analysis to cross informal passageways and parklands during estimation of travel paths to determine if this provides approximations of travel paths that more closely resemble actual travel paths. GPS offers accurate estimates of distance and barriers encountered, yet requires increased time commitments compared to self-report and GIS, therefore, the need for detailed data must be balanced with the requirements of the research.

This research, although innovative, had several limitations. First, only a single day of monitoring was used. We acknowledge that this may not accurately capture usual travel patterns and that future research using multiple travel days may be required to obtain a better understanding of children's active transport. Second, several journeys were removed due to signal dropout, which is an unavoidable limitation of GPS technology and again, monitoring over several days may increase the number of journeys available for analysis. We acknowledge that GPS signal dropout may occur due to heavy foliage over travel routes and interruption of transmission signals from satellites by multistory buildings. This may limit the widespread use of GPS, especially in urban environments containing many multistory buildings, as significant signal dropout may negatively affect the collection of complete journey data. The use of GPS should be explored in a variety of environments to determine the relative utility of the approach in a variety of circumstances. Finally, it is unknown if there was any reactivity to the measurement process and if the presence of the GPS unit altered normal travel patterns. Although pedometer-based research suggests that reactivity does not occur when the study purpose is described to participants,[9] there is still a need for future studies to address this potential limitation. Despite these limitations, this research is strengthened by comparisons made between barriers encountered and distances traveled measured by GIS and GPS.

Conclusion

Global Positioning Systems offer many benefits associated with measuring active transport compared with traditional methods of assessment, such as assessing actual travel distances and travel patterns in urban environments. Additionally, GPS more accurately reflects the frequency of barriers to active transport encountered than is estimated by GIS. Multiple days of monitoring may be required to overcome data loss associated with GPS signal dropout.

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References

1. Rodriguez D, Joo J. The relationship between non-motorized travel behaviour and the local physical environment. Transport Res D-Tr E 2004;9:151–73.
2. Timperio A, Ball K, Salmon J, et al. Personal, family, social, and environmental correlates of active commuting to school. Am J Prev Med 2006;30:45–51.
3. Boarnet MG, Anderson CL, Day K, McMillan T, Alfonzo M. Evaluation of the California safe routes to school legislation: urban form changes and children's active transportation to school. Am J Prev Med 2005;(2 Suppl 2):134–40.
4. Duncan MJ, Mummery WK. Reliability and concurrent validity of global positioning systems (GPS) for measuring distance traveled and route taken during active transportation. J Sci Med Sport 2005;8:S197.
5. Rodriguez D, Brown AL, Troped PJ. Portable global positioning units to complement accelerometry-based physical activity monitors. Med Sci Sports Exerc 2005;37(11 Suppl):S572–81.
6. Turner S, Shunk G, Hottenstein A. Development of a methodology to estimate bicycle and pedestrian travel demand: College Station: Texas Transportation Institute; 1998.
7. Duncan M, Mummery K. Psychosocial and environmental factors associated with physical activity among city dwellers in regional Queensland. Prev Med 2005;40:363–72.
8. Austroads. Urban Speed Management in Australia. Sydney: Austroads; 1996.
9. Ozdoba R, Corbin CB, Le Masurier GC. Does reactivity exist in children when measuring activity levels with unsealed pedometers? Ped Exer Sci 2004;16:158–66.